project performs an in-depth **Exploratory Data Analysis (EDA)** on a Telecom Customer Churn dataset to identify the major factors responsible for customer attrition. The analysis includes data preprocessing, handling missing values, feature transformation, and extensive visualization techniques to uncover hidden trends and relationships within the data. The primary objective is to understand customer behavior, identify high-risk customer segments, and provide actionable business recommendations that can help reduce churn and improve customer retention.

1. Overall Customer Churn

The analysis reveals that **26.54%** of customers have discontinued the telecom service, while **73.46%** have remained loyal.

- **Customers Retained: 73.46%**
- **Customers Churned: 26.54%**

This indicates that **approximately one out of every four customers leaves the company**, making churn a significant business concern. Although the majority of customers remain with the service, reducing even a small portion of the churn rate could substantially increase long-term revenue and customer lifetime value.

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2. Gender-wise Analysis

The customer base is almost equally distributed between male and female customers, and both groups exhibit very similar churn rates.

- Male Customers: ~**50%** of total customers
- Female Customers: ~**50%** of total customers

The visualizations indicate that **gender has little to no influence on customer churn**, suggesting that retention strategies should focus on behavioral and service-related factors rather than demographic characteristics like gender.

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3. Senior Citizen Analysis

Senior citizens account for a relatively smaller proportion of the customer base (around **16%**), while non-senior customers make up approximately **84%**.

However, despite their smaller population, **senior citizens exhibit a considerably higher churn rate** than younger customers. This suggests that elderly customers may face challenges related to service complexity, pricing, or customer support.

The findings indicate that telecom companies should introduce:

- Personalized customer assistance,
- Simplified service plans,
- Dedicated customer support,
- Senior-friendly promotional offers,

to improve customer satisfaction within this segment.

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4. Tenure Analysis

Customer tenure emerges as one of the strongest predictors of churn.

The analysis shows that customers with **shorter subscription periods (particularly within the first few months)** are significantly more likely to leave the company.

Conversely, customers with **higher tenure (more than one year)** demonstrate much greater loyalty and considerably lower churn rates.

This pattern suggests that:

- The **highest risk period occurs during the initial months after customer acquisition**.
- Customers who remain beyond this period are much more likely to continue using the service.

Businesses should therefore invest in:

- Better onboarding experiences,
- Welcome offers,
- Early customer engagement,
- Follow-up support during the first 90 days.

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5. Contract Type Analysis

Contract duration has one of the strongest relationships with customer churn.

The visualizations clearly indicate that:

- **Month-to-Month Contract customers experience the highest churn rate**, contributing to the majority of customer losses.
- Customers with **One-Year Contracts** have significantly lower churn.

- Customers enrolled in **Two-Year Contracts exhibit the lowest churn percentage**, demonstrating strong customer loyalty.

This trend suggests that long-term contractual commitments increase customer retention and reduce the likelihood of switching to competitors.

The company should encourage customers to migrate from month-to-month contracts to annual or two-year plans through:

- Discounted pricing,
- Loyalty rewards,
- Exclusive benefits,
- Bundled service packages.

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6. Internet and Additional Services

The analysis demonstrates that customers subscribing to additional value-added services are considerably less likely to churn.

Customers **without** the following services exhibit noticeably higher churn:

- Online Security
- Online Backup
- Device Protection
- Tech Support

These services enhance customer satisfaction by improving reliability, security, and technical assistance, making customers more likely to remain with the company.

Promoting these services during customer onboarding or through targeted marketing campaigns could significantly improve customer retention.

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7. Payment Method Analysis

Payment method also plays an important role in customer churn behavior.

Customers using **Electronic Check** display the **highest churn percentage** among all payment options.

In contrast, customers paying through:

- Automatic Bank Transfer
- Credit Card (Automatic Payment)

show considerably lower churn rates.

Automatic payment methods improve payment convenience, reduce missed payments, and increase customer commitment.

The company should incentivize customers to switch to automatic payment methods by offering:

- Cashback rewards,
- Discounted monthly bills,
- Loyalty points,
- Simplified payment setup.

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8. Monthly Charges and Total Charges

The analysis suggests that customers with **higher monthly charges** tend to churn more frequently, particularly when they are on month-to-month contracts.

Additionally, customers with **lower total charges** often correspond to newer customers, reinforcing the finding that early-stage customers are at the highest risk of leaving.

This indicates that pricing strategies and perceived value significantly influence customer retention.

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9. Business Recommendations

Based on the insights obtained from the analysis, the following strategic recommendations are proposed:

- Improve customer onboarding during the first three months to reduce early churn.
- Encourage migration from **Month-to-Month contracts** to **One-Year or Two-Year contracts** through discounts and loyalty incentives.
- Promote value-added services such as **Online Security, Device Protection, Online Backup, and Tech Support** to increase customer engagement.
- Design personalized retention campaigns specifically targeting senior citizens.
- Encourage customers to adopt automatic payment methods instead of electronic checks.
- Monitor customers with high monthly charges and provide customized plans or promotional offers to improve perceived value.
- Implement predictive churn monitoring to proactively identify high-risk customers and intervene before they leave.